

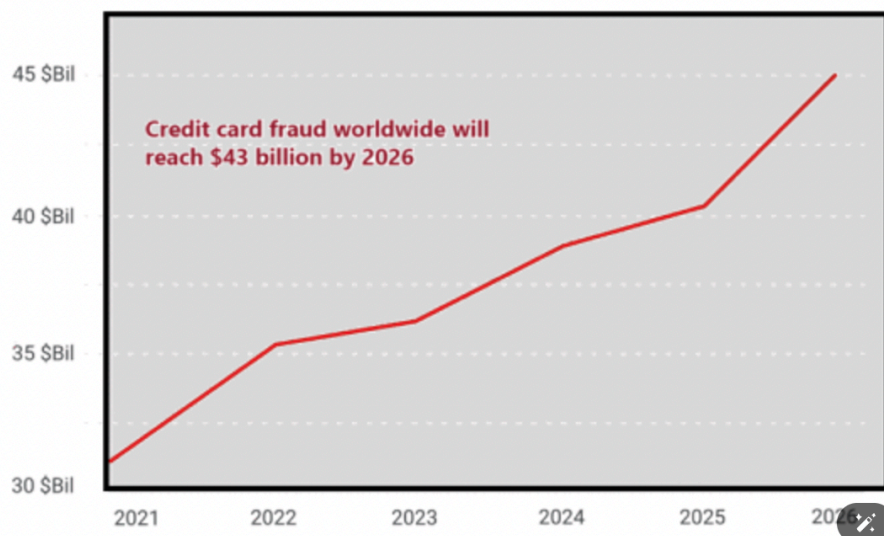
FRAUD DETECTION IN CREDIT CARD TRANSACTIONS USING MACHINE LEARNING

(By Sophie Blessing M, G Sai Chetan, N Swathi)

Over the last few years, the internet's advancement has led the technology and made a huge evolution of business. The root reason for the escalation in businesses today is around financial assessment. As per the present scenario, 100 percent of finance is done using the internet, i.e., online or cashless. The main sources of transactions are debit and credit card usage, online and mobile banking, and, not least, different transactional applications and websites. Due to the increasing use of transactional sources, there is a rapid increase in transactional fraud

Credit card fraud is a widespread financial crime that involves using someone else's credit card information without permission to make purchases or withdraw money. This type of fraud can happen through various techniques, including phishing, data breaches, skimming devices, and hacking. Criminals often use advanced methods to steal card details, which are used for unauthorized transactions or sold illegally. Victims can suffer financial losses and harm to their credit scores, while businesses face chargebacks and diminished customer trust. [Figure 1](#) shows an increase in credit card fraud instances and highlights how urgent it is to address this growing problem. The expansion of online shopping and digital payments has opened new avenues for credit card fraud. With more people making online purchases, there's an increased flow of sensitive card information that cybercriminals can exploit. E-commerce sites, though convenient, often attract hackers looking for security weaknesses to steal customer data. Phishing scams and fraudulent websites also trick consumers into revealing their credit card details. In response to

the rise in fraud, various solutions have been implemented by institutions like government and private banks, insurance companies, and cybersecurity firms. These entities have introduced crucial laws, regulations, and a range of policies. For instance, new account holders in banking are often encouraged to use additional services such as credit cards, internet banking, and mobile banking, all of which come with security measures like monthly statements and insurance against fraudulent activities. While these solutions provide some protection, the key to addressing fraud is accurately identifying the perpetrator and understanding the methods used. Advanced algorithms may play an important role in addressing the root cause of the challenge by examining all the factors involved.



Credit card fraud detection faces a notable challenge stemming from data imbalance, wherein fraudulent and non-fraudulent transactions are unevenly distributed within datasets. This disparity introduces the possibility of a skewed model and limited fraud detection efficacy. Previous research has addressed this issue by employing various methodologies such as data balancing, oversampling, under-sampling, and leveraging techniques like the

Synthetic Minority Oversampling Technique (SMOTE) to handle inconsistent credit card fraud data. Nonetheless, there is a distinct need for a thorough investigation into the effectiveness of these approaches to enhance the understanding and implementation of robust fraud detection mechanisms. Emerging advancements in machine learning (ML), particularly deep learning (DL) algorithms like Artificial Neural Networks (ANNs), Deep Boltzmann Machines, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Sequential models, play a crucial role in navigating toward accurate outcomes and identifying fraudulent transactions [1], [3], [4]. Credit card transaction irregularities are found and examined via regression analysis, hypothesis testing, and clustering. By analysing previously collected information, ML systems can detect forged transactions in real time. DL uses neural networks to identify complicated patterns and characteristics in large datasets, resulting in accurate fraud detection. Detecting credit card scams using a DL-based sequential model requires many critical components.

Data preprocessing is essential for cleaning and normalizing data and engineering features. The model architecture typically includes RNNs like LSTM or GRU, which capture temporal dependencies and may utilize embedding layers and attention mechanisms. Training the model requires suitable loss functions, optimization algorithms like Adam, and regularization techniques to prevent overfitting. Evaluation metrics that include precision, recall, F1 score, and ROC-AUC are crucial for performance assessment, especially given the class imbalance in fraud detection. Techniques like resampling, synthetic data generation and cost-sensitive learning address this imbalance. Deployment involves efficient inference pipelines, continuous performance monitoring, and post-processing for decision-making through threshold setting and risk scoring. Lastly, model explainability tools like SHAP or LIME are essential for understanding and

interpreting model decisions, integrating the model effectively into existing transaction processing systems, and incorporating feedback loops for continuous improvement.

Ensemble learning enhances the recognition of credit card fraud by integrating numerous approaches to utilize their distinct strengths while reducing individual weaknesses. This approach helps capture diverse patterns, improve accuracy, and generalize better to unseen data. By averaging predictions, ensemble methods mitigate overfitting and are more robust to noise. Techniques like boosting sequentially improve model performance while handling class imbalance more effectively. Overall, ensembles provide better-calibrated probabilities, increase prediction confidence, and reduce errors, resulting in a more accurate and reliable fraud detection system.

This study employs a Sequential model to optimize transactional fraud detection through a novel DL approach. As implied by its name, the model operates within the realm of neural networking, harnessing its capabilities to sift through complex transactional data. Various ensemble techniques such as logistic regression, random forest, Gradient Boost, and AdaBoost are initially utilized for training and accuracy assessment to increase the model's effectiveness. [Figure 2](#) shows the process of different ensemble learning models used in this paper. Among these, Gradient Boost emerges as the most accurate. Furthermore, the Sequential model is tailored with multiple layers, including dense, activation, and dropout layers, with a total of seven dense layers employed. Each layer is meticulously configured with distinct input parameters, utilizing Rectified Linear Unit (ReLU) activation functions and a sigmoid output layer. Through comparison and refinement, a Sequential model configuration yielding superior accuracy is developed, representing a significant advancement in fraud detection methodology. [Figure 3](#) shows the basic architecture of

the sequential model. Stochastic Gradient Descent (SGD) is a powerful and efficient optimizer for training ML models, particularly well-suited for large-scale datasets and DL. Its efficiency, ability to escape local minima, ease of implementation, adaptability with advanced variants, and implicit regularization make it a versatile and widely used optimization technique. By updating model parameters using only a single or small batch of examples, SGD speeds up computation and reduces memory usage. Fraudulent activities in credit card transactions often require immediate detection to prevent further losses. SGD's ability to update model parameters quickly makes it suitable for real-time fraud detection systems, enabling rapid response to emerging fraudulent patterns. [Figure 4](#) illustrates the architecture of the proposed system.

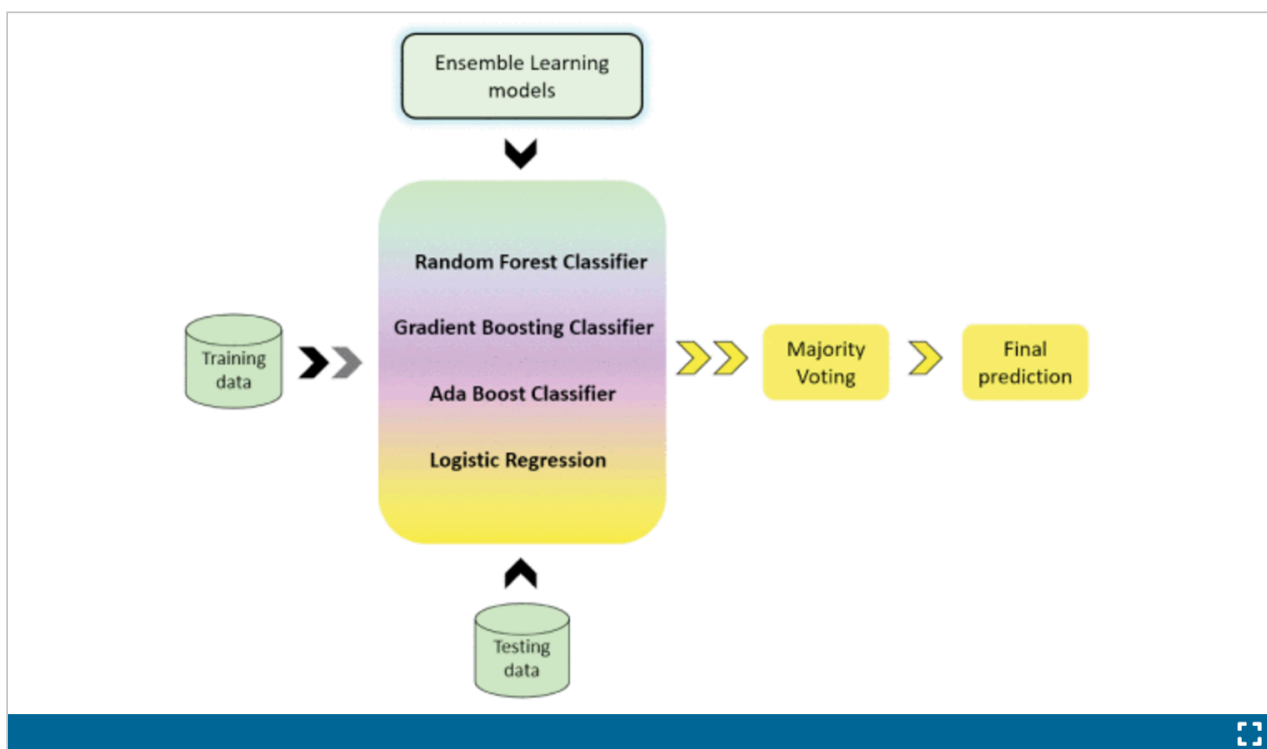


FIGURE 2.
Process of ensemble learning models.

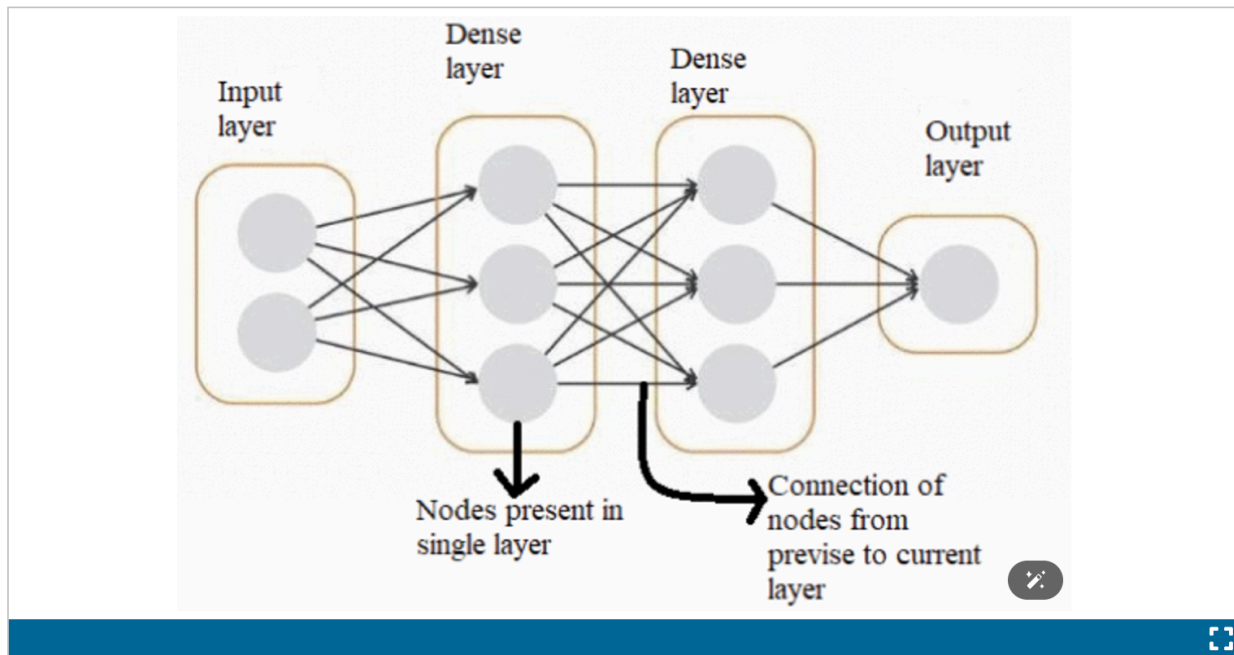


FIGURE 3.
Architecture of Sequential model.

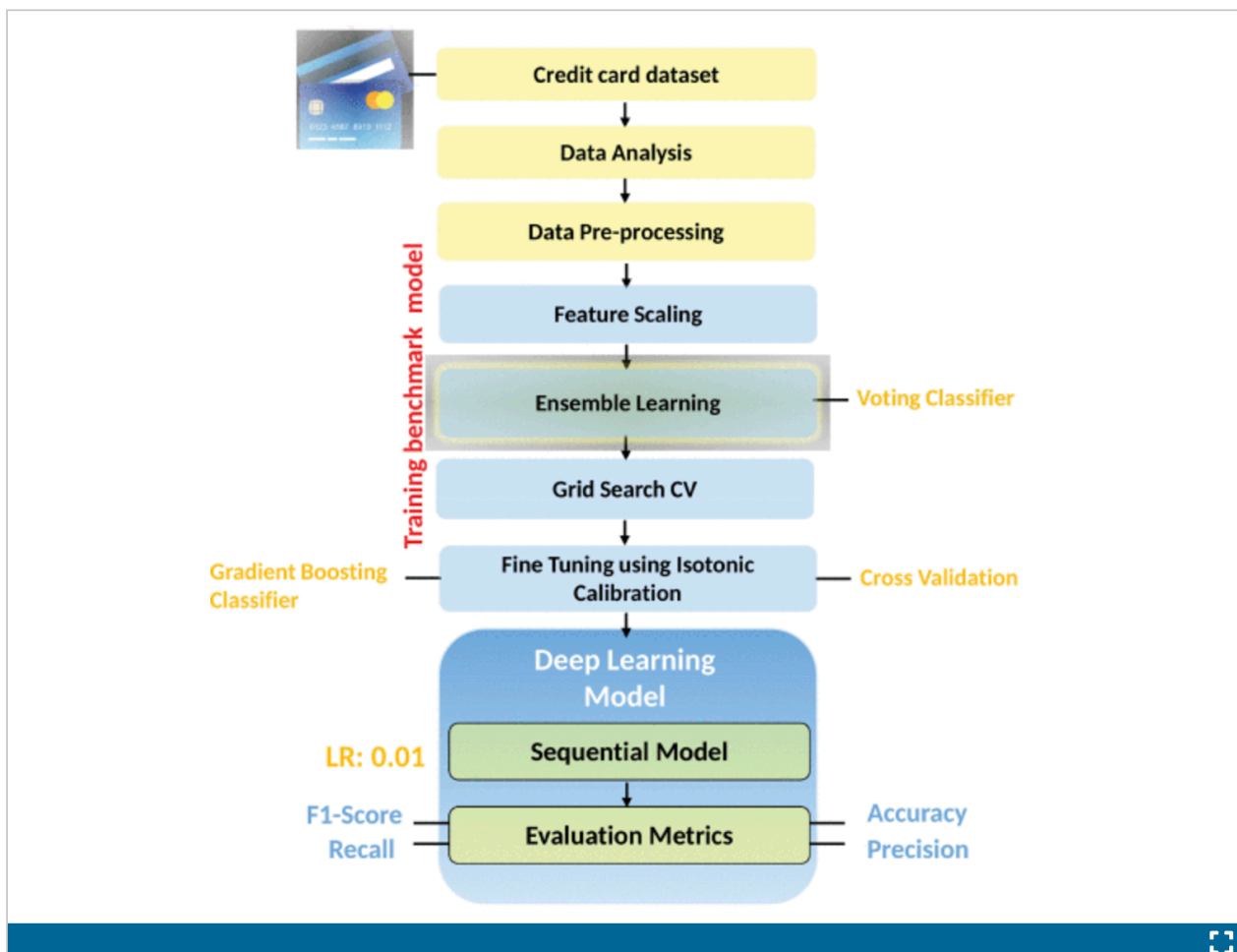


FIGURE 4.
Model architecture of the proposed system.

The following sections of this article are arranged as follows. [Sections II](#) and [III](#) focus on various reviews in detecting fraudulent transactions and digitalization in the finance sector. [Section IV](#) provides the dataset description and other DL model steps. [Section V](#) sheds light on creating the deep-learning model architecture and the techniques for fabricating the proposed model. [Section VI](#) illustrates the metrics for evaluating the framework, including accuracy, precision, recall, and F1-score. [Section VII](#) discusses the theoretical contributions and implications for practice. Ultimately, [Section VIII](#) describes the derived Conclusions.

SECTION II.

Related Work

Slowly but steadily, the financial spectrum has seen a significant transformation in its processes due to the emergence of digital technology. Technology's benefits have had an impact on the rate of Financial Globalization. As a result, the capacity to manage large operations while remaining responsive to any technological disruption is required. Financial technology businesses are revolutionizing the financial industry due to simple access to digital technologies, ushering in a new era of marketing strategies.

This paper reviews and describes inter-organizational development patterns in the banking industry's digital transformation [5]. Fintech is being introduced due to society's digital revolution for transactions, cash transfers, and other financial activity. Financial technology is being used and mistreated by criminals in the seedy underworld for fraud, intimidation, and black money. Fintech and virtual transaction

activity exploration should be acknowledged as an emerging technological sub-discipline within the digital forensics ecosystem.

Traditional banks are reluctant to fully embrace the new potential of the digitalized financial sector to the technology's qualities, and banks have postponed critical decisions to upgrade their technology [6]. The authors prioritize trust in the digital transformation of financial services, as it is a key foundation for all innovations and the long-term development of the financial sector in the digital economy [7]. Many pre-digital firms across conventional industries are concerned about developing and implementing a digital transformation strategy (DTS), but how such a plan may be formed remains a mystery. The authors point out that the digital approach to decision-making not only reflects an interruption in the traditions of upfront strategic information systems (IS) planning but also uncovers a new level of intrinsic decision formulation by focusing on the factors involved in European financial services providers and strategizing activities [8]. Customer involvement and digitalized services are two digital approaches that provide a point of reference for a digital transformation journey [9].

Transaction fraud has become a significant issue with the growing rise of digital financial services. Extracting users' behaviour profiles (BPs) from their prior transaction records and verifying whether an incoming transaction is a fraud based on their BPs is an effective way to identify fraud. A logical BP (LGBP) graph is proposed to represent the logical relationship of transaction record characteristics [10]. In online transactions, this study suggests a fraud detection model that works on convolutional neural networks, which builds an input feature sequencing layer that accomplishes the restructuring of dry activity characteristics to produce distinct convolutional trends [11]. TitAnt, a transaction

fraud detection platform used by Ant Financial, one of the world's top Fintech firms, is introduced to address the fraud issue. In milliseconds, the technology can predict online real-time transaction fraud [12].

Criminals use techniques such as Trojan or Phishing to illicitly acquire information from others' credit cards. Consequently, the need for robust fraud detection technology becomes crucial. This technology plays a vital role in identifying fraud in real-time, especially when a thief attempts to purchase using a stolen card. One approach involves leveraging ML techniques to analyze a comprehensive set of transaction data, encompassing both legitimate and fraudulent transactions. By establishing standard behavioral patterns based on this data, the system can assess whether a transaction is likely fraudulent. Two variants of random forests were used [13] to develop the action elements of valid and invalid transactions. First, historical transaction data splits all cardholders into groups with comparable transaction habits. The classifier is used to overcome the problem of idea drift and to detect fraud online. A feedback mechanism is used in the identification process if a new transaction is illegal. A window-sliding technique was suggested to aggregate the transactions in each group [14].

An approach to modeling individual behavior was introduced by leveraging the collective behavior of current transaction groups and their statuses, which enhanced the accuracy of low-frequency users' behavior prediction. This method involves a transaction behavior-based multi-behavior detection model. The Naive Bayes algorithm was employed to calculate the probability of a transaction being fraudulent based on the outcomes of various behaviors and, subsequently, to decide whether the transaction was fraudulent. The performance of user transactions declines due to the interaction of these three factors [15].

- The authors use historical transaction data and an optimal risk threshold strategy to establish a standard for evaluating transaction patterns.
- DBSCAN clustering is applied to extract behavior characteristics from normal and fraudulent transactions, creating a typical behavior profile for the transaction group.
- A sliding window method is employed to determine the current transaction status by leveraging information from previous transactions.

Using a real-world dataset, the authors deploy various supervised ML techniques to detect credit card fraud. These algorithms also create a super classifier with ensemble learning approaches [16]. To account for skewed datasets, this study employed a random under-sampling method using three different dataset proportions. The three ML techniques this study uses for credit card fraud detection are logistic regression, Naive Bayes, and K-nearest neighbor [17].

SECTION III.

The Profound Deep Learning

ML employs algorithms to understand and analyze data and make the best judgements based on those learnings. DL involves organizing algorithms into multiple layers to construct a neural network. While ML approaches are currently deployed to detect credit card fraud, achieving high performance has remained challenging for existing fraud detection systems. Recently, DL has emerged as a promising solution for addressing complex situations

in various fields [18]. ML methods have offered a considerable contribution to fraud detection mechanisms. However, to enhance the results obtained, DL approaches have successfully obtained better results.

The authors of this article proposed the BiLSTM-MaxPooling-BiGRU-MaxPooling model for detecting credit card fraud, which focuses on bidirectional Long Short-Term Memory (BiLSTM) and bidirectional Gated Recurrent Unit (BiGRU). The authors have also used ML classifiers (Naive Bayes, Random Forest, Ada Boost, Logistic Regression, Voting, and Decision Tree). Considering the DL and ML models, the result was that the DL model outperformed the other models [19]. The authors introduced a deep Autoencoder in the framework to achieve improved accuracy [20]. Between the effectiveness of DL approaches, a comparison was performed and various ML algorithms on three distinct financial datasets for credit card fraud detection. The results reveal that the proposed DL methods perform better than traditional ML models. These findings underscore the applicability of the diagnostic techniques in effectively detecting credit card fraud in real-world scenarios [21]. For detecting fraudulent transactions in real-time data streams, the authors offer an innovative framework called the Deep Ensemble Algorithm framework (DEAL). The suggested system is resilient to hidden transaction patterns, such as purchasing behavior, and adaptive to imbalanced data [22]. In this paper, the created model's performance is compared to the performance of two other standard ML algorithms (logistic regression and support vector machine). The findings reveal that the DL model outperforms the other two models regarding overall performance [23].

SECTION IV.

Materials and Methods

DL is at the heart of a slew of high-tech breakthroughs. Deep understanding is being used in various industries to ensure its success. Because most DL approaches rely on neural network topologies, DL models are also defined as deep neural networks. With a DL model, an algorithm can use its neural network to decide whether a prediction is correct.

A. Dataset Description

For the training and evaluation of our model, we employed the Credit Card Fraud Detection dataset [24]. This dataset encompasses transaction records from European credit cardholders over two days. This dataset identified 492 instances of fraudulent activity out of 284,807 transactions. The dataset is notably imbalanced, with fraudulent transactions constituting only 0.172% of the total. Each transaction includes 28 features, V1 through V28, and is anonymized using Principal Component Analysis (PCA) to address confidentiality concerns. The ‘Time’ and ‘Amount’ features were not subject to PCA; ‘Time’ indicates the seconds elapsed between each transaction and the first transaction in the dataset, while ‘Amount’ represents the transaction value. The response variable, ‘Class’, indicates fraud with a value of 1 and non-fraud with a value of 0. After selecting the dataset, the initial phase was to preprocess the data to ensure it was suitable for model training and testing. This involved identifying and addressing any missing values, standardizing the ‘Amount’ column for easier analysis, removing the ‘Time’ column due to its minimal contribution, and detecting and eliminating duplicate entries.

B. Data Analysis and Pre-Processing

Data preprocessing involves transforming raw data into a refined dataset by addressing tasks like data cleaning, integration, transformation, and reduction. It ensures the data is prepared for

training by algorithms. Preprocessing refers to the steps taken to modify data before using it in algorithms. These steps typically include importing libraries, reading data, handling missing values, encoding categorical data, standardizing data, performing PCA transformation, and splitting data. Different preprocessing techniques are in ML. In this credit card fraud detection analysis, feature engineering faced obstacles because the dataset lacked detailed feature information. This limitation prevented the application of feature selection techniques, which depend on clear feature details. This step was deliberately skipped to prevent any potential confusion from automated feature selection. Moreover, the 'Time' column was disregarded during manual analysis as it only represented a sequential count of entries, lacking any temporal relevance.

1) Feature Scaling

- Feature scaling is a strategy for condensing the data's independent features into a set range.
- It is used to handle significantly changing intensities and values during data pre-processing.

There are two most crucial ways of performing feature scaling, which are as follows:

- Min-Max Normalization
- Standardization

In this research, to carry out feature scaling, Min-Max Normalization is utilized. This method again scales the feature or observed value with a distribution value ranging from 0 to 1. The equation that contributes towards the calculation of feature scaling using Min-Max Normalization is as follows:

$$X_{scaled} = \frac{X_i - \min(X)}{\max(x) - \min(X)}$$

where X is the feature value after scaling, and $\min$ and $\max$ are the feature ranges. Sklearn in Python provides a way to scale features using Min-Max Normalization. After feature scaling, all the feature values will now be in the range of 0 to 1.

Next, we assess the distribution of class labels in the dataset. Upon evaluation, it was observed that 99.83% of the data samples belong to the legitimate class '0', while only 0.17% belong to

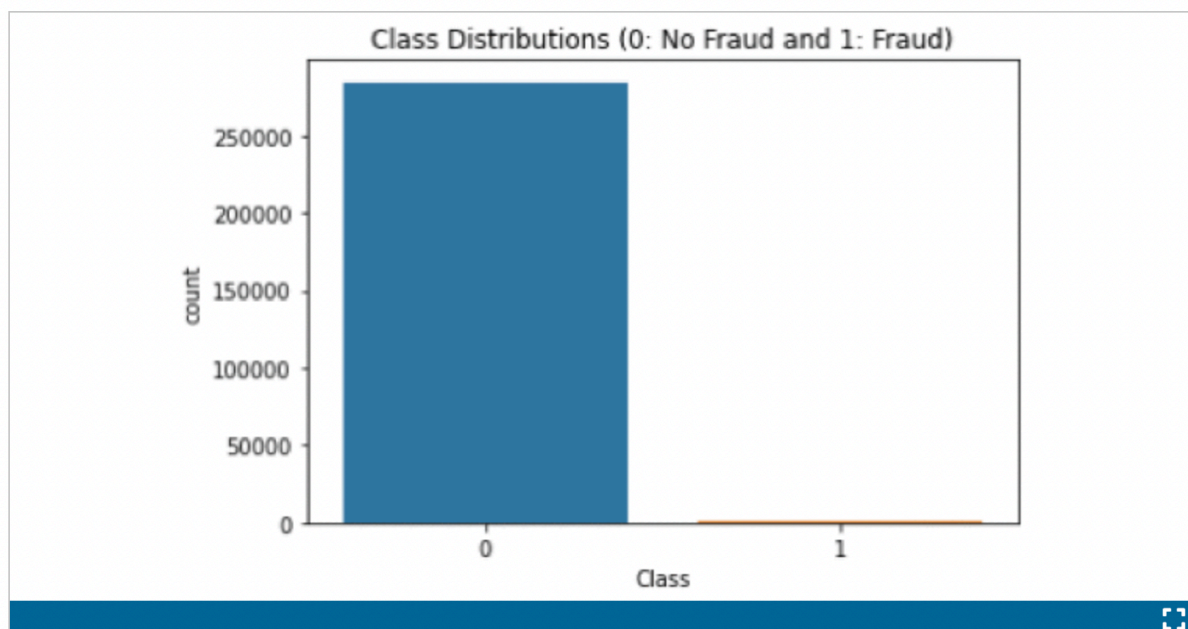


FIGURE 5.
Count of each class.

class '1'. This indicates a significant class imbalance. [Figure 5](#) illustrates the proportions of the majority and minority classes present in the dataset.

C. Data Sampling

Following pre-processing, the next phase addressed data imbalance through sampling techniques. Initially, the dataset consisted of 275,190 authentic records and 473 fraud-labelled records, highlighting an important skewness for model training. Under-sampling and SMOTE were the two sampling techniques used in this research. [Figure 6](#) shows the differences between the undersampling and oversampling processes. Under-sampling involves selecting a random subset from the majority class (legitimate dealings categorized as 0), with the number of samples determined to match the size of the minority class. The dataset was balanced by equating the number of legitimate entries to the smaller class records and concatenating both to form a single dataset. SMOTE is a statistical technique used to increase the number of smaller class occurrences in a balanced manner. It generates new instances from existing minority cases. In this research, the fraudulent transactions (marked as 1) were oversampled to balance the quantity with legitimate transactions. This method ensured an equal representation of both classes. After oversampling, the data from both classes were merged into one dataset to facilitate optimal model training.

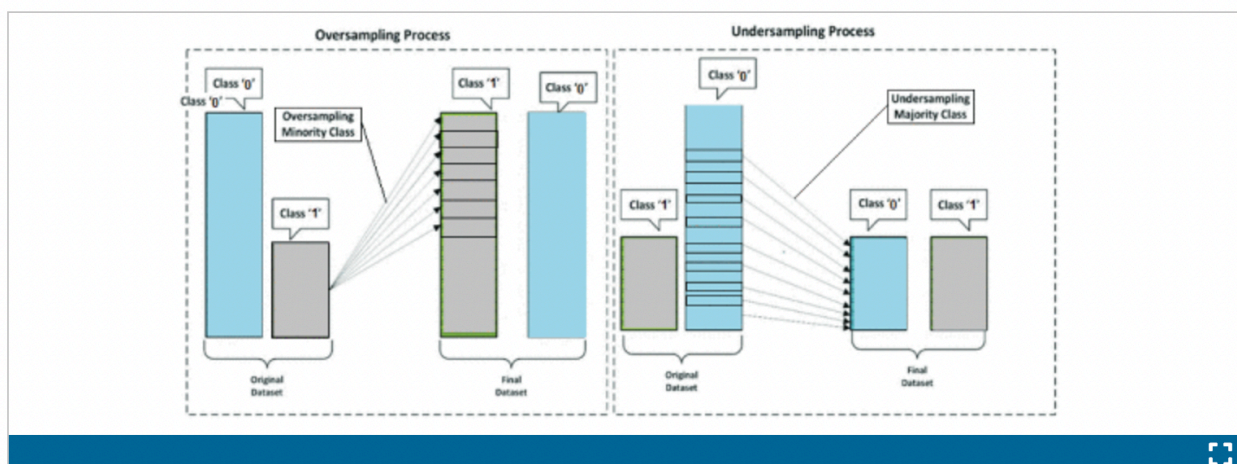


FIGURE 6.
Oversampling and undersampling process.

D. Initial Model Training

To avoid overfitting further, it is always a good practice to set aside some of the available data as a test set while running through the research project. Following the initial data sampling process, the dataset underwent partitioning into distinct subsets designated for training and testing purposes. The training subset was utilized to facilitate model training and subsequent evaluation of its performance, while the testing subset served to gauge the model's efficacy on novel data instances. Before partitioning, the dataset's 'Class' column, containing categorical labels, was isolated. Subsequently, the dataset was divided into two segments: one for training (comprising 80% of the data) and another for testing (representing 20%). Thereafter, the training subset underwent model training, a process primarily dedicated on discerning underlying patterns and correlations within the dataset. Employing selected ML algorithms entailed iterative learning and adaptation to the nuanced characteristics of credit card transactions. Model refinement ensued through the iterative adjustment of algorithmic parameters to enhance predictive accuracy. After training, the models underwent comprehensive evaluation utilizing the training dataset and earlier untouched testing data, offering a robust calculation of their generalizability and predictive capabilities.

E. Training the Benchmark Model - Ensemble Learning

Because the changes and prediction models do not match, the predicted results may be erroneous and sometimes even come out to be unreliable. Using an ensemble learning-based strategy, the authors in this article mention that once a situation changes, the prediction models can be trained

using historical data and ensemble learning, allowing a reliable and effective prediction model to be built based on the prediction models [25].

A comparative analysis of various ensemble learning methods was presented in [26]. It effectively examines the performance of Logistic Regression, Naive Bayes, K-Nearest Neighbors, Multi-Layer Perceptron, AdaBoost, Quadratic Discriminant Analysis, Random Forests, and the Pipelining method in detecting fraudulent credit card transactions. Among these, the Pipelining method demonstrated superior performance across different evaluation metrics.

Instead of relying on a single model as the ultimate predictor, ensemble approaches consider multiple models and aggregate their outputs to create a final model. Ensemble methods aim to enhance overall performance by combining insights from diverse models. Ensemble approaches are particularly effective in mitigating the negative impacts of bias, variance, and noise on a model’s predictions and errors.

TABLE 1 Different Metrics Achieved by Using Respective Classifiers

	Parameters			
Classifiers	Accuracy	Precision	Recall	f1_score
Logistic Regression	0.778	0.61	0.778	0.68
Random Forest Classifier	0.81	0.79	0.81	0.79
Gradient Boosting Classifier	0.82	0.8	0.82	0.79
Ada Boost Classifier	0.81	0.79	0.81	0.78
Voting Classifier	0.81	0.79	0.81	0.79
Tuned Gradient Boosting Classifier	0.82	0.81	0.82	0.8

As shown in [Table 1](#) the accuracy, precision, recall and F1-score for the classifiers in ensemble learning are mentioned. To start training the benchmark model, the following classifiers are used.

Logistic Regression:

- Logistic Regression is one of the finest ML algorithms. It is used for solving classification problems.
- It is a predictive analysis-oriented algorithm based on the probability theory. We have used logistic regression to train the model to find its accuracy and compared the model using a sequential DL model.

Random Forest Classifier:

- Another ensemble algorithm that uses the bagging technique is Random Forest.
- Random forest is regarded as an ensemble model since it is a group of decision trees that have been combined to produce a more accurate model.

Gradient Boosting Classifier:

- Gradient Tree Boosting is a boosting that applies to any variational loss function.

Ada Boost Classifier:

- The core concept of AdaBoost is to sequentially train a series of weak classifiers on progressively adjusted versions of the data.

- In each iteration of boosting, weights $w_1, w_2, \dots, w_N$

1

, w_2

2

, ..., w_N

N

are assigned to the training samples, where each weight w_i

i

is initially set to $1/N$

. Consequently, the initial step involves training a weak classifier on the original data without modifications.

Voting Classifier:

- It is one of the basic ensembling techniques. The maximum voting approach is commonly utilized for classification problems.
- Multiple models are utilized in this strategy to produce predictions for every piece of data. In this research, 'VotingClassifier,' a module in Sklearn provided by Python for maximum voting mechanism, is used.

1) Grid Search CV and Cross Validation

Grid Search with Cross-Validation (GridSearch CV) uses the Cross-Validation method to test all possible combinations of the values specified in the database and assesses the model for each one. As a result, after using this function, the accuracy and loss can be determined for each combination of hyperparameters, and select the one that performs the best. By utilizing Grid Search CV, we can methodically and effectively adjust the hyperparameters of a model, ensuring it performs well on new data. This ensures that the model performs well on new data, which is essential in credit card fraud detection, as the impact of false negatives (overlooking fake transactions) can be significant.

2) Calibration

While implementing classification, it's common to know the class label and probability of that identifier. This probability gives some assurance that the prediction is correct. Some models can provide inaccurate forecasts of class probabilities, and many don't support predicting probabilities. The calibration module enables fine-tuning a model's probabilities or even incorporating the functionality

of probability prediction. Calibrating a classifier entails trying to fit a calibrator that assigns the classifier's output to a calibrated probability in between the range [0, 1]. The equation of calibration can be denoted as:

$$C = p(y_i = 1|f_i)$$

where f is the output of the classifier.

The leveraged calibration module significantly contributed to the enhancement of the credit card fraud detection model. This module played a pivotal role in optimizing the model's parameters to improve its accuracy in identifying fraudulent transactions. Initially trained on a dataset comprising both legitimate and fraudulent credit card transactions, the model required refinement to ensure its efficacy in real-world scenarios. Calibration was essential to minimize the occurrences of false positives and false negatives by fine-tuning the model's classification thresholds. This iterative process involved analyzing the model's predictions against actual outcomes, identifying discrepancies, and adjusting decision boundaries accordingly. Through these iterative adjustments, the calibration module effectively aligned the model's responses with the characteristics of fraudulent transactions while reducing misclassifications.

T